CUSTOMER PURCHASE ANALYTICS

PROJECT REPORT

Python • Pandas • SQL • Power BI

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1. Executive Summary

Customer Purchase Analytics is an end-to-end portfolio project that explores customer shopping records to understand revenue distribution, category and product performance, seasonal patterns, customer purchase behavior, ratings, and subscription status. It combines Python/Pandas for data inspection and preparation, SQL for structured analysis, and Power BI for interactive reporting.

The final dashboard displays Total Revenue, Number of Customers, Average Purchase Amount, and Average Review Rating, plus visuals for revenue by category and season, top products by revenue, subscription status, and revenue by season and location.

2. Problem Statement

Retail and e-commerce teams need a practical way to explore purchase records, identify the categories and products contributing most to revenue, and examine differences across seasons, locations, and customer segments. This project prepares the data, queries key metrics, and presents selected results in an interactive dashboard.

3. Objectives

- Inspect dataset structure, data types, missing values, duplicate rows, and descriptive statistics.
- Standardize text, remove exact duplicate rows, convert selected fields to numeric values, and fill missing review ratings with the median.
- Create derived segments for age, purchase value, customer value, discount status, repeat-buyer status, ratings, and purchase frequency.
- Use SQL to calculate counts, revenue, averages, category and seasonal performance, customer spending, and product rankings.
- Build a Power BI dashboard to communicate headline metrics and category, product, season, location, and subscription patterns.

4. Tools and Technologies

Tool	Role
Python / Jupyter Notebook	Data inspection, cleaning, feature engineering, exploratory analysis, aggregation, and export
Pandas and NumPy	Data manipulation, grouping, segmentation, and missing-value handling
Matplotlib	Exploratory charts
SQL / MySQL	Aggregation, filtering, subqueries, CASE WHEN, CTEs, and window functions
Power BI Desktop	KPI cards, interactive visuals, slicers, and dashboard presentation

5. Dataset Overview

The notebook reads a CSV named customer_shopping_behavior.csv. Fields used include Customer ID, Age, Gender, Category, Item Purchased, Purchase Amount (USD), Review Rating, Previous Purchases, Discount Applied, Frequency of Purchases, Season, Payment Method, and customer/subscription attributes.

The SQL script expects database customer_purchase_analytics and table customer_behavior_clean. The notebook exports customer_behavior_clean.csv. Ensure paths, schema, and column names are aligned before reproducing the project.

The dashboard screenshot displays approximately $233K total revenue, $59.76 average purchase amount, a 3.75 average review rating, and 3.9K for the card labelled Number of Customers. These are screenshot values and were not independently recalculated for this report.

6. Methodology

6.1 Data inspection

The notebook previews records, checks row and column counts, lists columns, inspects data types and non-null values, calculates descriptive statistics, checks missing values and exact duplicates, and examines value counts for selected categorical fields.

6.2 Data cleaning

- Trim leading and trailing whitespace from text columns.
- Remove exact duplicate rows.
- Convert Purchase Amount (USD), Review Rating, and Previous Purchases to numeric values, coercing invalid values to missing.
- Fill missing Review Rating values with the median rating.

The notebook prints the cleaned dataset shape after these steps. Results should be verified by running the notebook; this report does not claim that every possible data-quality issue has been resolved.

6.3 Feature engineering

Derived field	Logic in notebook
Age Group	18–24, 25–34, 35–44, 45–54, 55–64, and 65+
Purchase Band	Low up to $40; Medium above $40 through $70; High above $70
Customer Value Segment	High Value when Previous Purchases ≥35; Growing when ≥15; otherwise New / Low Value
Discount Flag	Discounted if Discount Applied is 'yes' after lowercase conversion; otherwise Full Price
Repeat Buyer Flag	Repeat if Previous Purchases >0; otherwise First Purchase
Rating Band	Needs Attention up to 3; Positive above 3 through 4; Excellent above 4
Frequency Group	Weekly/Bi-Weekly/Fortnightly = High; Monthly/Quarterly/Every 3 Months = Medium; Annually = Low; otherwise Other

6.4 Exploratory analysis

The notebook calculates revenue, average purchase, row count, and unique Customer ID count; groups revenue by category; compares average purchase by gender; identifies frequently purchased products; summarises customers and categories; builds a category-by-season pivot; compares discount groups; reviews purchase-amount outliers using IQR; checks correlations among selected numeric fields; and exports analysis ready data.

7. SQL Analysis

The SQL script contains 25 numbered analytical queries, progressing from basic aggregation to intermediate analysis.

Technique	Examples
Basic aggregation	SELECT/LIMIT, COUNT, COUNT DISTINCT, SUM, AVG, MIN, MAX
Grouped analysis	Orders and revenue by category, average purchase by gender, average rating by category, revenue by season
Product/payment analysis	Top products by purchase count and most-used payment methods
Filtering/subqueries	Purchases above the overall average purchase amount
CASE WHEN / HAVING	Customer value segments, categories with at least 100 records, discounted versus full-price grouping
Customer-level analysis	Spending per customer, average order value, repeat versus one-time customer grouping
CTEs / window functions	Revenue share, RANK of customers, top products per category using PARTITION BY
Cross-dimensional analysis	Category-by-season revenue and high-rated products with minimum purchase volume

Execution requires the database and table to exist and the cleaned data to be loaded with matching column names. MySQL 8.0 or later is recommended for window-function support.

8. Power BI Dashboard

The final dashboard uses a dark navy background and blue accents. It combines KPI cards, bar/column charts, a line-and-column seasonal visual, a donut chart, an area chart, and slicers.

Component	Purpose / displayed result
Total Revenue	Approximately $233K; sum of purchase amounts
Number of Customers	3.9K displayed; confirm whether this counts rows or distinct Customer IDs
Average Purchase Amount	$59.76 displayed
Average Review Rating	3.75 displayed
Revenue by Category	Compares Clothing, Accessories, Footwear, and Outerwear
Revenue by Season	Compares seasonal revenue and a customer-count series
Top 10 Products by Revenue	Ranks products by summed purchase amount
Customers by Subscription Status	Compares Yes and No subscription categories
Revenue by Season & Location	Explores revenue across season and location
Slicers	Subscription Status, Gender, Category, and Shipping Type visible in screenshot

Use Sum of Purchase Amount (USD) for revenue, Average of Purchase Amount (USD) for average purchase, Average of Review Rating for ratings, and Distinct Count of Customer ID when the intended metric is unique customers. Verify the Number of Customers card: the displayed 3.9K may be a record count rather than a distinct-customer count.

9. Key Observations

- Clothing is the highest-revenue category in the dashboard, at approximately 104K.
- Accessories follows at approximately 74K, then Footwear at approximately 36K and Outerwear at approximately 19K.
- The dashboard displays approximately $233K total revenue, $59.76 average purchase amount, and 3.75 average review rating.
- The Top 10 Products by Revenue visual highlights high-revenue items.
- The subscription donut compares Yes and No subscription statuses.
- The season-and-location visual compares those dimensions; it is not a true date-based trend because the listed dataset fields do not include a genuine order date.

10. Business Recommendations

- Maintain visibility into Clothing, the strongest revenue category shown, while monitoring its margins and inventory availability.
- Investigate lower-revenue categories such as Outerwear using product mix, purchase frequency, discounts, and ratings before changing inventory or marketing spend.
- Use the Top 10 products view to inform merchandising discussions, validating margins and stock constraints before prioritising products.
- Compare seasonal and location patterns for campaign and assortment planning, while checking sample sizes and data definitions.
- Compare subscriber and non-subscriber purchasing patterns to identify potential subscription opportunities.
- Validate the definitions of customer count, transaction count, and repeat customer before using these metrics for decisions.

11. Limitations and Validation Notes

- No genuine order-date field is established by the supplied notebook. A Jan–Dec revenue trend should not be invented without date data.
- The 3.9K KPI is labelled Number of Customers, but the notebook distinguishes row count from unique Customer ID count. Confirm the card aggregation.
- The SQL script uses Discount Applied; the notebook derives Discount Flag from that source field. Confirm schema consistency across Python, SQL, and Power BI.
- The notebook reads customer_shopping_behavior.csv and exports customer_behavior_clean.csv. Document the input dataset source and provide the expected file.
- Notebook cells and SQL queries were not executed for this report; the dashboard KPI values are transcribed from the supplied screenshot.
- Credit the original dataset source and follow its licence; do not claim a third-party dataset as original.

12. Conclusion

Customer Purchase Analytics demonstrates an end-to-end analytics workflow combining data-quality checks, feature engineering, exploratory analysis, SQL aggregation and ranking, and interactive Power BI reporting. The dashboard presents headline metrics and category, product, season, location, rating, and subscription patterns. The project will be stronger after validating the customer-count KPI, documenting dataset provenance, confirming the PBIX and CSV are included, and recording reproducible execution results.

Appendix: Interview Talking Points

- Python was used for inspection, cleaning, feature engineering, and exploratory analysis.
- SQL was used for repeatable aggregations, filtering, subqueries, CTEs, and ranking with window functions.
- Power BI was used to make the results interactive and easy for stakeholders to explore.
- The main displayed category finding is that Clothing leads revenue.
- Possible next improvements include validating unique-customer counts, adding real date data if trend analysis is needed, and documenting the source and refresh steps.